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Class FileSystemElement

- java.lang.Object
 - FileSystemElement
- Direct Known Subclasses:
 - [Directory](#), [File](#)

```
public abstract class FileSystemElement
extends java.lang.Object
```

- ◦ **Field Summary**

Fields		
Modifier and Type	Field	Description
protected java.sql.Timestamp	dateCreated	
protected java.lang.String	name	
protected FileSystemElement	parent	

- **Constructor Summary**

Constructors		Description
Constructor		
FileSystemElement	(java.lang.String name, FileSystemElement parent)	

- **Method Summary**

All Methods Instance Methods Concrete Methods		
Modifier and Type	Method	Description
java.sql.Timestamp	getDateCreated()	
java.lang.String	getName()	
FileSystemElement	getParent()	
void	setName (java.lang.String name)	
FileSystemElement	setParent (FileSystemElement parent)	

- **Methods inherited from class java.lang.Object**

clone, equals, finalize, getClass, hashCode, notify, notifyAll, toString, wait, wait, wait

- ◦ **Field Detail**

- **name**

protected java.lang.String name

- **dateCreated**

protected java.sql.Timestamp dateCreated

- **parent**

protected [FileSystemElement](#) parent

- **Constructor Detail**

- **FileSystemElement**

```
public FileSystemElement(java.lang.String name,  
                          FileSystemElement parent)
```

- **Method Detail**

- **getName**

```
public java.lang.String getName()
```

- **getDateCreated**

```
public java.sql.Timestamp getDateCreated()
```

- **getParent**

```
public FileSystemElement getParent()
```

- **setName**

```
public void setName(java.lang.String name)
```

- **setParent**

```
public FileSystemElement setParent(FileSystemElement parent)
```

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Class File

- java.lang.Object
- - [FileSystemElement](#)
 - - File

- public class File
extends [FileSystemElement](#)

- - **Field Summary**

- **Fields inherited from class [FileSystemElement](#)**

[dateCreated](#), [name](#), [parent](#)

- - **Constructor Summary**

Constructors

Constructor	Description
File (java.lang.String name, FileSystemElement parent)	

- - **Method Summary**

- **Methods inherited from class [FileSystemElement](#)**

[getDateCreated](#), [getName](#), [getParent](#), [setName](#), [setParent](#)

- **Methods inherited from class java.lang.Object**

[clone](#), [equals](#), [finalize](#), [getClass](#), [hashCode](#), [notify](#), [notifyAll](#), [toString](#), [wait](#), [wait](#), [wait](#)

- - **Constructor Detail**

- **File**

```
public File(java.lang.String name,  
            FileSystemElement parent)
```

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Class Directory

- java.lang.Object
 - [FileSystemElement](#)
 - Directory

- public class Directory
 - extends [FileSystemElement](#)

- ◦ **Field Summary**

- **Fields inherited from class [FileSystemElement](#)**

[dateCreated](#), [name](#), [parent](#)

- **Constructor Summary**

Constructors

Constructor	Description
Directory (java.lang.String name, FileSystemElement parent)	

- **Method Summary**

All Methods [Instance Methods](#) [Concrete Methods](#)

Modifier and Type	Method	Description
void	addChild (FileSystemElement child)	
java.util.LinkedList< FileSystemElement >	getChildren ()	
void	removeChild (FileSystemElement child)	

- **Methods inherited from class [FileSystemElement](#)**

[getDateCreated](#), [getName](#), [getParent](#), [setName](#), [setParent](#)

- **Methods inherited from class java.lang.Object**

[clone](#), [equals](#), [finalize](#), [getClass](#), [hashCode](#), [notify](#), [notifyAll](#), [toString](#), [wait](#), [wait](#), [wait](#)

- ◦ **Constructor Detail**

- **Directory**

public Directory(java.lang.String name,
[FileSystemElement](#) parent)

- **Method Detail**

- **getChildren**

- ```
public java.util.LinkedList<FileSystemElement> getChildren()
```

- **addChild**

- ```
public void addChild(FileSystemElement child)
```

- **removeChild**

- ```
public void removeChild(FileSystemElement child)
```

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## Class FileSystem

- java.lang.Object
  - FileSystem

- public class FileSystem
  - extends java.lang.Object

- ◦ **Constructor Summary**

Constructors

### Constructor Description

[FileSystem\(\)](#)

- **Method Summary**

All Methods [Instance Methods](#) [Concrete Methods](#)

| Modifier and Type         | Method                                                                                                                                                                          | Description                                                                                                                                                      |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">Directory</a> | <a href="#">ChangeDirectory</a> ( <a href="#">Directory</a> currentDirectory, java.lang.String newPath)                                                                         | Changing the current directory by first taking the newPath and splitting it with '/' to an array then to ArrayList for removing and simplicity.                  |
| void                      | <a href="#">CreateFileorDirectory</a> ( <a href="#">Directory</a> currentDirectory, java.lang.String currentPath)                                                               | Creating a new directory or file after taking the type and name from the user.                                                                                   |
| void                      | <a href="#">DeleteFileorDirectory</a> ( <a href="#">Directory</a> currentDirectory, java.lang.String fileName, java.util.LinkedList< <a href="#">FileSystemElement</a> > child) | Using the children of the current directory we are first checking if the first child is the file to be deleted if it is we delete it and set its parent to null. |
| java.lang.String          | <a href="#">getRemainingPath</a> (java.util.ArrayList<java.lang.String> names)                                                                                                  | Helper function to get the remaining path and return it for changing the directory mehods and etc.                                                               |
| <a href="#">Directory</a> | <a href="#">getRoot</a> ()                                                                                                                                                      |                                                                                                                                                                  |
| void                      | <a href="#">ListContents</a> ( <a href="#">Directory</a> currentDirectory, java.lang.String currentPath)                                                                        | Listing the contents of the current directory using an enhanced loop it checks if it is a file or directory using 'instanceof' then prints it accordingly.       |
| void                      | <a href="#">MoveFileorDirectory</a> ( <a href="#">Directory</a> currentDirectory, java.lang.String currentPath)                                                                 | Moving the directory by first finding the file to be moved then removing the file to be moved from the current directory with index remove.                      |
|                           | <a href="#">PrintDirectoryTree</a> ( <a href="#">Directory</a> currentDirectory,                                                                                                | Printing the directory tree by using our current path to iterate and finally printing the files and                                                              |

|         |                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|---------|-----------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| void    | <code>java.lang.String currentPath)</code>                                                                                              | directories of our current directory regarding to their types using '*' for directories and also printing ' ' as in the example output.                                                                                                                                                                                                                                                                                                                |
| boolean | <code><a href="#">SearchForFiles</a>(<a href="#">Directory</a> dir,<br/>java.lang.String searchQuery,<br/>java.lang.String path)</code> | Searching the given file in our file management system by first checking if the file is in the children directory if it is we are printing the path If it is not when we find a <code>FileSystemElement</code> which is an instance of <code>Directory</code> then We call the method recursively by changing the current directory to that <code>Directory</code> and it continues until All the directories have been searched or the file is found. |
| void    | <code><a href="#">SortContents</a>(<a href="#">Directory</a> currentDirectory)</code>                                                   | Sorting the contents of the current directory by first taking the size and creating a temp array to assign the children to here for sorting.                                                                                                                                                                                                                                                                                                           |

## ▪ Methods inherited from class `java.lang.Object`

`clone, equals, finalize, getClass, hashCode, notify, notifyAll, toString, wait, wait, wait`

## • ◦ Constructor Detail

### ▪ `FileSystem`

`public FileSystem()`

## ◦ Method Detail

### ▪ `getRoot`

`public Directory getRoot()`

### ▪ `getRemainingPath`

`public java.lang.String getRemainingPath(java.util.ArrayList<java.lang.String> names)`

Helper function to get the remaining path and return it for changing the directory methods and etc.

Parameters:

names -

Returns:

### ▪ `ChangeDirectory`

`public Directory ChangeDirectory(Directory currentDirectory,  
java.lang.String newPath)`

Changing the current directory by first taking the `newPath` and splitting it with '/' to an array then to `ArrayList` for removing and simplicity. Then going to the first directory in the path by checking the name of the current child directory and if it is found changing our current directory and calling the method again with our new `currentDirectory` location it continues till the last directory in the path to go to.

Parameters:

currentDirectory -

newPath -

Returns:

Last directory in the path.

### ▪ `ListContents`

`public void ListContents(Directory currentDirectory,  
java.lang.String currentPath)`

Listing the contents of the current directory using an enhanced loop it checks if it is a file or directory using 'instanceof' then prints it accordingly.

Parameters:

currentDirectory -

currentPath -

### ▪ `CreateFileorDirectory`

`public void CreateFileorDirectory(Directory currentDirectory,  
java.lang.String currentPath)`

Creating a new directory or file after taking the type and name from the user. According to the type ;  
Creating the new file/directory using the add method of directory and setting its parent to our current directory using the constructor of File.

Parameters:

currentDirectory -  
currentPath -

## ■ DeleteFileorDirectory

```
public void DeleteFileorDirectory(Directory currentDirectory,
 java.lang.String fileName,
 java.util.LinkedList<FileSystemElement> child)
```

Using the children of the current directory we are first checking if the first child is the file to be deleted if it is we delete it and set its parent to null. If not we create a new linked list for children which does not consist of the already visited child which is the first one then calling our method recursively by using this new children linked list until we find the file to be deleted or size of the linked list is 1.

Parameters:

currentDirectory -  
fileName -  
child -

## ■ MoveFileorDirectory

```
public void MoveFileorDirectory(Directory currentDirectory,
 java.lang.String currentPath)
```

Moving the directory by first finding the file to be moved then removing the file to be moved from the current directory with index remove. After that, using our change directory method we change our current directory to the path that the file will be moved. After going to the path the file is going to be moved we add the file to be moved to that directory and printing if it was successful.

Parameters:

currentDirectory -  
currentPath -

## ■ SearchForFiles

```
public boolean SearchForFiles(Directory dir,
 java.lang.String searchQuery,
 java.lang.String path)
```

Searching the given file in our file management system by first checking if the file is in the children directory if it is we are printing the path If it is not when we find a FileSystemElement which is an instance of Directory then We call the method recursively by changing the current directory to that Directory and it continues until All the directories have been searched or the file is found.

Parameters:

dir -  
searchQuery -  
path -

## ■ PrintDirectoryTree

```
public void PrintDirectoryTree(Directory currentDirectory,
 java.lang.String currentPath)
```

Printing the directory tree by using our current path to iterate and finally printing the files and directories of our current directory regarding to their types using '\*' for directories and also printing ' ' as in the example output.

Parameters:

currentDirectory -  
currentPath -

## ■ SortContents

```
public void SortContents(Directory currentDirectory)
```

Sorting the contents of the current directory by first taking the size and creating a temp array to assign the children to here for sorting. Then, in a loop it sorts the array which includes the children using bubble sort by using before method of timestamp finally it prints the sorted array.

Parameters:

currentDirectory -

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## Class TheMenu

- java.lang.Object
- ◦ TheMenu

- ```
public class TheMenu
    extends java.lang.Object
```

- ◦ **Constructor Summary**

Constructors

Constructor Description

[TheMenu\(\)](#)

- ◦ **Method Summary**

All Methods [Static Methods](#) [Concrete Methods](#)

Modifier and Type Method	Description
static void main (java.lang.String[] args)	
static void TheMenu ()	Menu method for the user to manage the file system.

- **Methods inherited from class java.lang.Object**

clone, equals, finalize, getClass, hashCode, notify, notifyAll, toString, wait, wait, wait

- ◦ **Constructor Detail**

- **TheMenu**

public TheMenu()

- ◦ **Method Detail**

- **main**

public static void main(java.lang.String[] args)

- **TheMenu**

public static void TheMenu()

Menu method for the user to manage the file system.

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